

Evaluation Methods and Metrics

Exercise 6, May 25, 2021

1 Learning Objectives

- Understand ways of evaluating the “goodness” of a recommendation, and of a recommender algorithm or system.
- Understand how predictions and recommendations (including top-n) are evaluated.

2 Overview

During this exercise you will explore and understand results provided by several evaluation methods for Recommender Systems. If you need more information regarding the metrics, please refer to the bibliography at the end of this document.

Disclaimer: The code used for this exercise has been adapted from
<https://github.com/statisticianinstilettos/recmetrics>

To achieve the objectives of this exercise, comparisons in the recommendation performance between three models will be performed:

- Random recommender, recommends 10 random movies to each user.
- Popularity recommender, recommends the top 10 most popular movies to each user.
- Collaborative Filter, matrix factorization approach using single vector decomposition-SVD.

3 Instructions

In Moodle, you will find two Python files: `evaluation_metrics.py` (main program) and `metrics_lib.py` (library with functions to compute the metrics and display some plots) please download them, install the dependencies that you might be missing, and proceed to follow the next steps.

1. Download the MovieLens Dataset (`ml-20m.zip`) from <https://grouplens.org/datasets/movielens/20m/>. From this folder you will only need the files `movies.csv` and `ratings.csv`. Please fix the code according to the location of the downloaded files on `evaluation_metrics.py`, lines 9 and 24, so the main program is able to run.

2. **Implementation of the Recommender Models:** The models described in the Overview section are implemented from the lines 45 to 101 of the main program
3. **Running the code:** When you run the code some graphs and results will appear in your console, please read the document before continuing. Keep in mind that this dataset is large and it might take some seconds to show results on your console.

Some of the first results you will see are some lines of the two CSV files and also the predictions of movies ID that the users might like. The column `cf_predictions` contains a list of the movies ID predicted by means of collaborative filtering; `pop_predictions` and `random_predictions` are results of prediction by means of the popular and random recommender respectively
4. **Long tail plot - Lines 38 to 43:** This plot is used to explore popularity patterns in user-item interaction data such as clicks, ratings, or purchases. Typically, only a small percentage of items have a high volume of interactions, and this is referred to as the “head”. Most items are in the “long tail”, but they only make up a small percentage of interactions. Observe the graph generated when running the code and answer the following questions:
 - How will you describe the 1482 items that are in the head of the graph?
 - How will you describe the 20534 items that are in the tail of the graph?
 - If a RS recommends items that are only in the head of the graph, what can you say about this RS in terms of personalization and serendipity?
5. **MEAN AVERAGE RECALL @ K - Lines 126 - 151:** A recommender system typically produces an ordered list of recommendations for each user in the test set. MAR@K gives insight into how well the recommender is able to recall all the items the user has rated positively in the test set. Observe and analyze the MAR@K graph and answer the following:
 - Given that the CF model provides better MAR@K values, do you think this model will provide better recommendations than the random and popular models?
 - With this new information, which model do you think will perform better in terms of personalization of recommendations and serendipity?
6. **Coverage - Lines 134 - 151 :** Coverage is the percent of items in the training data the model is able to recommend. Observe and analyze the Coverage graph and answer the following:
 - Are you surprised that the random recommender shows a higher coverage than the other two models? Please explain why/why not.
 - With this new information, which model do you think will perform better in terms of personalization of recommendations and serendipity?
7. **Novelty - Lines 154 - 160 :** Novelty measures the capacity of recommender system to propose novel and unexpected items which a user is unlikely to know about already. You will find the results of the novelty measure on the console. Answer the following:
 - Which model do you think will perform better in terms of personalization of recommendations and serendipity?

8. **Conclusions:** Now that you have insights about the data (long tail plot) and also some performance indicators of the recommendations:

- Which model will you define as the overall best?
- When you think exclusively in terms of personalization and serendepity, which one do you consider the best model?
- If you have to pick one to implement in a real-life movies recommendation application, which model would you select to implement and why?

References

- [1] Schröder, Gunnar, Maik Thiele, and Wolfgang Lehner. "Setting goals and choosing metrics for recommender system evaluations." UCERSTI2 workshop at the 5th ACM conference on recommender systems, Chicago, USA. Vol. 23. 2011. URL: <https://wiki.epfl.ch/edicpublic/documents/Candidacy%20exam/Evaluation.pdf>
- [2] Shani, Guy, and Asela Gunawardana. "Evaluating recommendation systems." Recommender systems handbook. Springer, Boston, MA, 2011. 257-297. URL: <http://www.bgu.ac.il/~shanigu/Publications/EvaluationMetrics.17.pdf>